Network design and configuration
CAT 1

Question 1 (12 Marks)

Karatina University has hired you as a network admin to manage their network. The management has provided you with an IP address range of 10.0.0.0/24 to work with. They require multiple departments to have their own subnets for better organization and security. Using VLSM design a subnetting scheme for the company that meets the departmental requirements as shown in the table below.

Department	**Number of devices**
ICT Lab 6	100 desktops, 2 laptops and 2 printers
Exams Office	8 Computers and 1 printer
All Faculty Offices	51 omputers and 1 shared printer

a) Provide the following details for each subnet:

i. Network Address
ii. Usable IP Range
iii. Broadcast Address
iv. Subnet Mask

Question Two: (20 Marks)

The Open Shortest Path First (OSPF) routing protocol is a fundamental component of computer networks. Imagine you are a network administrator overseeing OSPF implementation in your organization.

Required:

i. Define the purpose of OSPF in computer networks. (2 Marks)
ii. Discuss the step-by-step process of how OSPF creates the routing information table. (6 Marks)
iii. Discuss the concept of OSPF areas and elaborate on their significance in the OSPF routing protocol. (4 Marks)
iv. Discuss how OSPF contributes to determining the shortest path to a destination within a computer network. (4 Marks)
v. Discuss four types of OSPF protocol packets that implement different functions in interaction between OSPF routers. (4 Marks)